

Nayan Bhatia

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github.com/nayanbhatia311 | [Google Scholar](https://scholar.google.com/citations?user=...)

Education

University of California, Santa Cruz Sep 2021 – Jun 2027 (exp.)
PhD Candidate in Computer Science & Engineering (transitioned from M.S)
[Inter-Networking Research Group \(i-NRG Lab\)](#) under Prof. Katia Obraczka
K.J. Somaiya Institute of Engineering and IT, University of Mumbai Aug 2017 – Jun 2021
B.E. in Computer Engineering

Experience

University of California, Santa Cruz Oct 2021 – Present
LLM & Platform Engineer

- Developed the **UC Basic Needs (UCBN) platform** (ucbasicneeds.ucsc.edu), a MERN-stack directory serving ~300,000 students across 10 UC campuses.
- Built and deployed the React (Vite + Tailwind) frontend, Express/MongoDB API, and Cloudflare-secured hosting with monitoring and analytics.
- Designed and maintained the **UC Essential Needs Navigator** (ucbasicneeds.ucsc.edu/ucen), an AI-powered search assistant enabling natural language queries with cited results.
- Scaled backend infrastructure with PM2, reverse proxies, and containerized CI/CD pipelines; improved uptime, SSL/TLS security, and performance.
- Instrumented site analytics with Cloudflare Web Analytics to monitor usage, load performance, and Core Web Vitals (~132 ms load time, all metrics in green).

Research & Projects

WiFiGPT: LLM-based Indoor Localization from Wi-Fi Telemetry 2025 – Present

- Developed **WiFiGPT**, a decoder-only transformer that treats Wi-Fi telemetry (CSI, RSSI, FTM) as a “language” and directly predicts device position.
- Reduced reliance on handcrafted feature engineering and improved cross-environment generalization.
- Achieved sub-meter accuracy on RSSI/FTM and centimeter-level precision on CSI across controlled testbeds.

Pulse-Fi: Contactless Vital-Signs via Commodity Wi-Fi 2024 – Present

- Built a low-cost sensing pipeline on ESP32 and Raspberry Pi devices to extract Wi-Fi Channel State Information (CSI) and estimate multiple vital signs.
- Demonstrated accurate monitoring of **heart rate, breathing rate, and fall detection** using ultra-low-cost commodity hardware.
- Co-led dataset curation and modeling across internal and public benchmarks; validated system against medical-grade sensors.
- Featured in notable outlets including **University of California News**, **Tom’s Hardware**, **Yahoo Tech**, and **CNET**, underscoring both academic credibility and global tech media recognition. Media coverage summary: [Pulse-Fi Press Coverage](#).

Publications & Preprints

- N. S. Bhatia, K. Obraczka. *Transforming Decoder-Only Transformers for Accurate WiFi-Telemetry Based Indoor Localization*. arXiv (2025): [Link](#). Patent Disclosure: [UC Case 2026-577-0](#).
- P. Kocheta, N. S. Bhatia, K. Obraczka. *Pulse-Fi: A Low-Cost System for Accurate Heart Rate Monitoring Using Wi-Fi Channel State Information*. In **IoT DCOSS (International Conference on Distributed Computing in Sensor Systems)**, Short paper/technical brief, 2025. [PDF](#).
- H. S. Kuttivelil, L. Krishnaswamy, N. S. Bhatia, K. Obraczka. **IEEE ICDCS 2025 Tutorial**: [Link](#).

- H. S. Kuttivelil, S. Sreenivasamurthy, L. Krishnaswamy, **N. S. Bhatia**, K. Obraczka. *Network Simulation Bridge: Bridging Applications to Network Simulators*. **ACM Q2SWinet**, 2023. [PDF](#).

Awards & Leadership

- **2026 QB3 Santa Cruz Graduate Innovators Fellowship**: Awarded for advancing PulseFi, a WiFi-based vital signs monitoring system with commercialization potential in healthcare.
- **CITRIS Smart Health Masterclass & Intensive (2025)**: Nominated for a 5-day program at UC Berkeley on responsible AI in healthcare, covering digital health strategy, policy, and healthtech innovation.
- **Certificate of Recognition (2024 and 2025)**: Awarded by UC Basic Needs & Economic Justice Center for exceptional contributions and leadership in advancing student resource initiatives).
- **Biodiversity Seed & Agroecology Fellowship**: \$11,000 award for UC Essential Needs (UCBN) platform.
- **SIP Mentor (2024 and 2025)**: Led a team of 4 interns on ML-based indoor localization; ~95%+ test accuracy in lab setting.
- **Young Gladiators, Colosseum Master Class (DARPA/NSF)**: Invited for a 3-day training for PhD/postdoc researchers to use the Colosseum wireless emulator for large-scale, hardware-in-the-loop experiments in Wi-Fi, cellular, ad hoc & multi-hop networking.
- **Teaching Assistant**: Supported instruction for various courses (Python, C, IoT, Database Systems, Operating systems) at UCSC; mentored 100+ students per quarter.
- **Peer Reviewer, IEEE Transactions on Mobile Computing (TMC)**: Reviewed a manuscript submission in the area of mobile and wireless systems.

Technologies

Programming: Python, SQL, C/C++, Java, Scala, Solidity

ML/AI: PyTorch, TensorFlow, Keras, NumPy/Pandas, scikit-learn, Matplotlib

Web: Node.js, React.js, Express, GraphQL, Flask, Django; HTML/CSS/JS

Data/Infra: MySQL, MongoDB, CUDA; Docker, Kubernetes, Nginx, GitHub Actions, CI/CD; AWS/GCP

MLOps/Automation: Jenkins, Ansible, Pytest/PyTorch-Lightning, Selenium, SLURM